

# Logistic Regression Master Guide (Telugu Transliteration)

Ee document lo, "LogisticsRegressionNotes.pdf" lo unna concepts ni base chesukuni, Classification problems ki use ayye **Logistic Regression** gurinchi pin-to-pin Telugu transliteration lo (English letters tho) chala detailed ga explain chestanu. Meeru nidra lo lechina error lekunda classification code rayagalige antha strong coding practice, math, mariyu visualizations ni ikkada include chesanu.

## 1. Logistic Regression Parichayam (Classification Ante Enti?)

### Logistic Regression ante enti?

Linear Regression numbers (continuous values) ni predict chesthe, **Logistic Regression** classifications ni predict chestundi. Idi oka **Supervised Machine Learning algorithm**, danini mukhyamga **Binary Classification** problems kosam vadataru.

- Binary Classification:** Output ki rendu classes matrame untayi (e.g., Yes/No, 0/1, Spam/Not Spam, Pass/Fail).
- Idi oka input givven class ki chendadaniki unna **probability (0 nunchi 1 madhya value)** ni predict chestundi.

### Nija Jeevitham lo Upayogalu (Real-Life Applications)

- Email Spam Detection:** E-mail Spam aa? Leda Inbox message aa?
- Medical Diagnosis:** Patient ki disease unda? Leda?
- Credit Card Fraud Detection:** Transaction Fraudulent aa? Genuine aa?
- Customer Churn:** Customer company nunchi vellipothara? Undipothara?

## 2. The Math Behind: Sigmoid Function

Linear Regression lo line straight ga  $-\infty$  nunchi  $+\infty$  varaku veltundi. Kani probability eppudu 0 mariyu 1 madhya ne undali. Anduke Logistic Regression lo oka special mathematical function vadataru — daanine **Sigmoid Function (Logistic Function)** antaru.

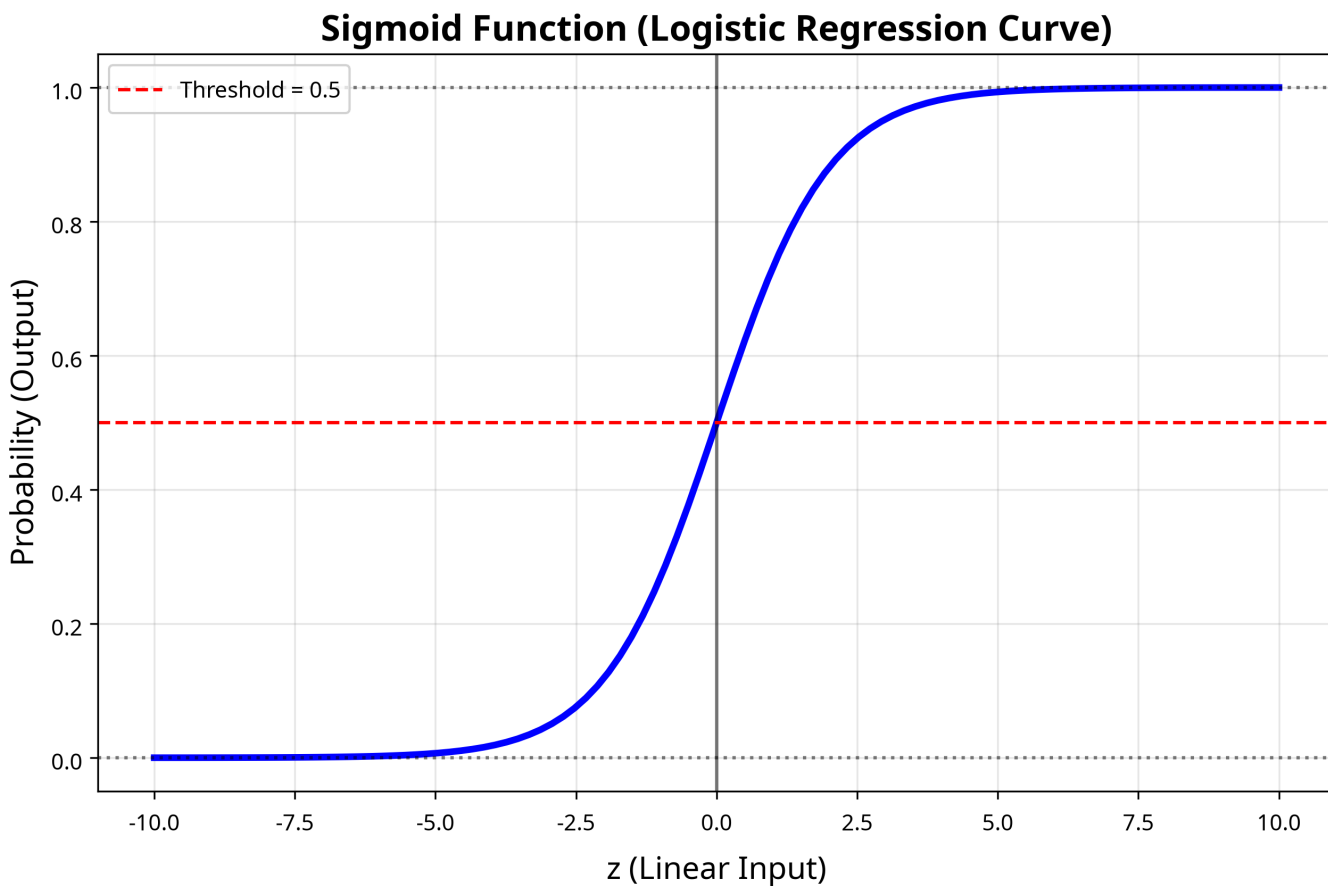
## Sigmoid Formula

$$\sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}}$$

Ikkada  $\mathbf{z}$  ante Linear Regression equation:  $\mathbf{z} = \mathbf{w}_1\mathbf{x}_1 + \mathbf{w}_2\mathbf{x}_2 + \dots + \mathbf{b} = \mathbf{W}^T\mathbf{X} + \mathbf{b}$

## Sigmoid Curve Visualization

Sigmoid function input (  $\mathbf{z}$  ) entha peddadaaina, output ni exact ga 0 mariyu 1 madhya ki compress chestundi. S-shaped curve la untundi:



## Decision Boundary (Garaapa Geetha)

Sigmoid output vachchin tarwata, model oka **Threshold** (saadharanamga **0.5**) ni set chestundi:

- Probability  $\geq 0.5$  ayithe  $\rightarrow$  Class 1 (e.g., Spam).
- Probability  $< 0.5$  ayithe  $\rightarrow$  Class 0 (e.g., Not Spam).

### 3. Classification Model Evaluation Metrics

Regression lo MAE, MSE vadinattu, Classification lo model entha accurate ga predict chesindo chudadaniki special metrics vadataru.

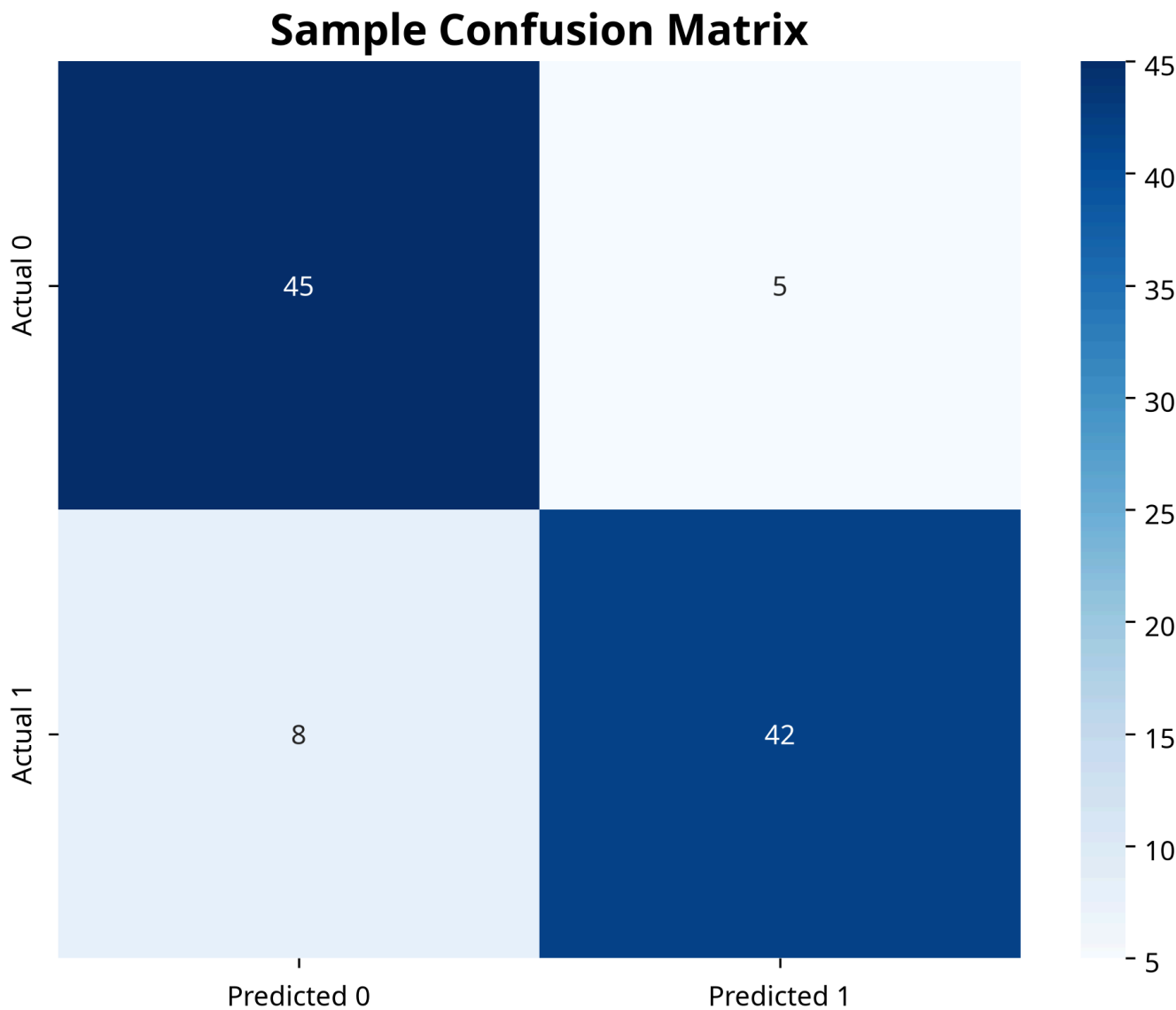
#### A. Confusion Matrix

Model predictions ni actual values tho compare chese oka table.

|                     | Predicted Negative (0)              | Predicted Positive (1)             |
|---------------------|-------------------------------------|------------------------------------|
| Actual Negative (0) | TN (True Negative)                  | FP (False Positive / Type I Error) |
| Actual Positive (1) | FN (False Negative / Type II Error) | TP (True Positive)                 |

- **TP (True Positive):** Actual ga positive, model kuda positive ani predict chesindi (Correct!).
- **TN (True Negative):** Actual ga negative, model kuda negative ani predict chesindi (Correct!).
- **FP (False Positive):** Actual ga negative, kani model positive ani cheppindi (Type I Error).
- **FN (False Negative):** Actual ga positive, kani model negative ani cheppindi (Type II Error).

Confusion Matrix Heatmap



B. Key Performance Metrics

- 1. **Accuracy:** Moktham predictions lo enni correct ga cheppam.  $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$
- 2. **Precision:** Model positive ani predict chesinvatillo enni nijamga positive unnayio. (False Positives takkuva undali anukunappudu idi mukhyam).  $Precision = \frac{TP}{TP+FP}$
- 3. **Recall (Sensitivity):** Actual positive cases lo model enni correct ga gurthupattindi. (False Negatives takkuva undali anukunappudu - e.g., Disease detection - idi mukhyam).  $Recall = \frac{TP}{TP+FN}$
- 4. **F1 Score:** Precision mariyu Recall yokka harmonic mean. Imbalanced datasets ki idi best metric.  $F1\ Score = 2 \times \frac{Precision \times Recall}{Precision+Recall}$

---

## 4. MASTER CODING SECTION: Step-by-Step Python Implementation

Meeru error lekunda code rayadaniki, scikit-learn use chesi Logistic Regression model train chesi, metrics calculate chese complete Python code ikkada ichchanu.

```

# =====
# STEP 1: Import Required Libraries
# =====
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score

# =====
# STEP 2: Create / Load Dataset
# =====
# Manam oka sample classification dataset create chestunnamu (e.g., Pass/Fail)
np.random.seed(42)
study_hours = np.random.uniform(1, 10, 100)
attendance = np.random.uniform(50, 100, 100)

# Target: Pass (1) if study_hours > 5 and attendance > 75, else Fail (0)
pass_fail = ((study_hours > 5) & (attendance > 75)).astype(int)

df = pd.DataFrame({'StudyHours': study_hours, 'Attendance': attendance, 'Passed': pass_fail})

# =====
# STEP 3: Define X (Features) and Y (Target)
# =====
X = df[['StudyHours', 'Attendance']]
Y = df['Passed']

# =====
# STEP 4: Train-Test Split
# =====
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

# =====
# STEP 5: Train Logistic Regression Model
# =====
model = LogisticRegression()
model.fit(X_train, Y_train)

# Predictions
Y_pred = model.predict(X_test)

# =====

```

```
# STEP 6: Evaluate Model (Metrics)
# =====
acc = accuracy_score(Y_test, Y_pred)
prec = precision_score(Y_test, Y_pred)
rec = recall_score(Y_test, Y_pred)
f1 = f1_score(Y_test, Y_pred)
cm = confusion_matrix(Y_test, Y_pred)

print("--- LOGISTIC REGRESSION EVALUATION ---")
print(f"Accuracy: {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall: {rec:.4f}")
print(f"F1 Score: {f1:.4f}")
print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(Y_test, Y_pred))
```

## 5. Real-Life & Data Science Applications

- **Email Filtering:** Spam vs Non-spam emails classification.
- **Credit Risk Analysis:** Loan applicant ki loan approve cheyacha leda (Default vs Non-default).
- **Healthcare:** Cancer detection (Malignant vs Benign tumors).

### AI/ML/DL Model Recommendations

Logistic Regression binary classification ki standard starting model. Multiclass classification ki **Softmax Regression (Multinomial Logistic Regression)** vadataru. Deep Learning lo binary classification kosam **Binary Cross-Entropy Loss** tho patu final layer lo **Sigmoid activation function** use chestaru.

### References:

1. [LogisticsRegressionNotes.pdf](#) - Original PDF document for Logistic Regression and classification metrics.